

Day 3

April 14, 2026

Production and Distribution

Bench, Line, Cell

DCs

Production and Distribution

- What type of factory do you use?
- How much capital?
- How much labor?
- What batch size?
- How do you distribute to retail customers?
- How do you distribute to your own DCs?
- How do you distribute to wholesale customer DCs?
- Do you use mail or container?

Production Technologies

Technology	A	α	β	Technology Description
Benches	0.009	0.10	0.85	Lab benches with general-purpose equipment. Each table is manned by a skilled worker who produces an entire product. Setup time for new batches = 0.05 days
Production Line	0.010	0.30	0.75	A row of tables with specialized equipment and each worker assigned a specific portion of the production sequence. Workers are relatively unskilled. Setup time for new batches = 0.50 days
Automated Cell	0.020	0.80	0.30	A robot that feeds material to automated production machines clustered around it. There is support staff but no workers performing actual production. Setup time for new batches = 1 day

Minimum Capital and Labor Requirements

Factory Type	Minimum Capital Investment	Minimum Daily Labor Expenditure
Bench	\$1	\$1 (\$0 to shut down)
Production Line	\$500,000	
Automated Cell	\$3,000,000	

NEVER put \$1 or anything close to \$1 for labor – you will ruin your factory.

Production Model

Daily Throughput (100 unit batches in this case)

		Bench						
		Daily Throughput in units, including setup time						
K ↓ vs L_daily →		500	1,000	2,500	3,000	4,000	8,000	16,000
100,000		2.310	4.161	9.044	10.552	13.455	24.123	43.065
200,000		2.476	4.459	9.690	11.305	14.414	25.832	46.085
300,000		2.578	4.643	10.089	11.770	15.006	26.887	47.946
400,000		2.654	4.778	10.382	12.112	15.441	27.661	49.311
500,000		2.713	4.886	10.615	12.383	15.786	28.276	50.396
600,000		2.763	4.975	10.809	12.610	16.075	28.789	51.299
700,000		2.806	5.052	10.976	12.804	16.322	29.229	52.075
800,000		2.844	5.120	11.123	12.975	16.540	29.616	52.757
900,000		2.877	5.181	11.254	13.128	16.734	29.962	53.365
1,000,000		2.908	5.235	11.372	13.266	16.910	30.275	53.915
2,000,000		3.116	5.610	12.184	14.211	18.113	32.412	57.673
3,000,000		3.245	5.841	12.685	14.795	18.855	33.731	59.988
4,000,000		3.340	6.011	13.052	15.224	19.400	34.698	61.685
5,000,000		3.415	6.147	13.345	15.564	19.833	35.467	63.033

		Line						
		Daily Throughput in units, including setup time						
K ↓ vs L_daily →		500	1,000	2,500	3,000	4,000	8,000	16,000
100,000		na	na	na	na	na	na	na
200,000		na	na	na	na	na	na	na
300,000		na	na	na	na	na	na	na
400,000		na	na	na	na	na	na	na
500,000		11.682	18.894	34.357	38.424	45.569	66.333	90.984
600,000		12.298	19.851	35.941	40.151	47.522	68.780	93.703
700,000		12.843	20.693	37.325	41.655	49.218	70.881	96.009
800,000		13.333	21.448	38.556	42.992	50.719	72.725	98.010
900,000		13.779	22.134	39.668	44.196	52.068	74.368	99.776
1,000,000		14.190	22.764	40.683	45.294	53.295	75.850	101.357
2,000,000		17.188	27.307	47.837	52.990	61.807	85.856	111.700
3,000,000		19.198	30.304	52.404	57.861	67.118	91.861	117.649
4,000,000		20.749	32.591	55.809	61.473	71.020	96.160	121.796
5,000,000		22.028	34.458	58.542	64.360	74.115	99.505	124.960

		Cell						
		Daily Throughput in units, including setup time						
K ↓ vs L_daily →		500	1,000	2,500	3,000	4,000	8,000	16,000
100,000		na	na	na	na	na	na	na
200,000		na	na	na	na	na	na	na
300,000		na	na	na	na	na	na	na
400,000		na	na	na	na	na	na	na
500,000		na	na	na	na	na	na	na
600,000		na	na	na	na	na	na	na
700,000		na	na	na	na	na	na	na
800,000		na	na	na	na	na	na	na
900,000		na	na	na	na	na	na	na
1,000,000		na	na	na	na	na	na	na
2,000,000		na	na	na	na	na	na	na
3,000,000		75.960	79.550	83.662	84.396	85.499	87.892	89.937
4,000,000		79.909	83.041	86.570	87.193	88.126	90.136	91.837
5,000,000		82.623	85.409	88.513	89.058	89.871	91.613	93.079

Production Model

Manufacturing Overhead per Unit

K ↓ vs L_daily →	Bench						
	Manufacturing Overhead per unit						
	500	1,000	2,500	3,000	4,000	8,000	16,000
100,000	224.33	244.74	278.45	286.04	298.64	332.39	371.96
200,000	216.72	232.49	261.78	268.61	280.05	311.11	347.98
300,000	215.23	227.22	253.24	259.55	270.22	299.59	334.85
400,000	216.03	224.63	247.86	253.75	263.80	291.87	325.96
500,000	218.02	223.43	244.15	249.66	259.18	286.16	319.31
600,000	220.72	223.08	241.45	246.63	255.68	281.70	314.04
700,000	223.87	223.31	239.45	244.31	252.92	278.08	309.71
800,000	227.35	223.93	237.94	242.50	250.70	275.07	306.06
900,000	231.05	224.85	236.79	241.08	248.88	272.51	302.91
1,000,000	234.93	226.00	235.94	239.95	247.38	270.30	300.16
2,000,000	278.00	243.55	235.26	236.87	241.06	258.12	283.78
3,000,000	323.41	265.25	240.41	239.91	241.29	253.46	275.88
4,000,000	369.10	288.22	247.66	245.18	243.95	251.67	271.26
5,000,000	414.60	311.67	255.96	251.58	247.85	251.38	268.36

K ↓ vs L_daily →	Line						
	Manufacturing Overhead per unit						
	500	1,000	2,500	3,000	4,000	8,000	16,000
100,000	na	na	na	na	na	na	na
200,000	na	na	na	na	na	na	na
300,000	na	na	na	na	na	na	na
400,000	na	na	na	na	na	na	na
500,000	50.64	57.77	75.43	80.46	89.79	121.98	176.86
600,000	49.59	55.91	72.62	77.46	86.48	117.91	171.93
700,000	48.92	54.52	70.41	75.10	83.88	114.67	167.99
800,000	48.49	53.46	68.64	73.19	81.75	112.02	164.74
900,000	48.25	52.63	67.18	71.61	79.99	109.79	162.01
1,000,000	48.14	51.97	65.95	70.28	78.49	107.89	159.67
2,000,000	50.40	50.03	59.92	63.53	70.64	97.45	146.52
3,000,000	54.66	51.13	58.19	61.34	67.78	93.07	140.67
4,000,000	59.40	53.16	57.92	60.72	66.64	90.81	137.38
5,000,000	64.27	55.60	58.35	60.84	66.33	89.60	135.37

K ↓ vs L_daily →	Cell						
	Manufacturing Overhead per unit						
	500	1,000	2,500	3,000	4,000	8,000	16,000
100,000	na	na	na	na	na	na	na
200,000	na	na	na	na	na	na	na
300,000	na	na	na	na	na	na	na
400,000	na	na	na	na	na	na	na
500,000	na	na	na	na	na	na	na
600,000	na	na	na	na	na	na	na
700,000	na	na	na	na	na	na	na
800,000	na	na	na	na	na	na	na
900,000	na	na	na	na	na	na	na
1,000,000	na	na	na	na	na	na	na
2,000,000	na	na	na	na	na	na	na
3,000,000	13.82	19.48	36.45	42.06	53.21	97.27	184.01
4,000,000	15.43	20.86	37.34	42.81	53.70	96.88	182.20
5,000,000	17.14	22.43	38.59	43.97	54.70	97.32	181.74

Production Model

Manufacturing Overhead + Materials Cost per Unit (\$100 in this case)

		Bench						
		Manufacturing Overhead plus Materials Cost per unit						
K ↓ vs L_daily →		500	1,000	2,500	3,000	4,000	8,000	16,000
100,000		324.33	344.74	378.45	386.04	398.64	432.39	471.96
200,000		316.72	332.49	361.78	368.61	380.05	411.11	447.98
300,000		315.23	327.22	353.24	359.55	370.22	399.59	434.85
400,000		316.03	324.63	347.86	353.75	363.80	391.87	425.96
500,000		318.02	323.43	344.15	349.66	359.18	386.16	419.31
600,000		320.72	323.08	341.45	346.63	355.68	381.70	414.04
700,000		323.87	323.31	339.45	344.31	352.92	378.08	409.71
800,000		327.35	323.93	337.94	342.50	350.70	375.07	406.06
900,000		331.05	324.85	336.79	341.08	348.88	372.51	402.91
1,000,000		334.93	326.00	335.94	339.95	347.38	370.30	400.16
2,000,000		378.00	343.55	335.26	336.87	341.06	358.12	383.78
3,000,000		423.41	365.25	340.41	339.91	341.29	353.46	375.88
4,000,000		469.10	388.22	347.66	345.18	343.95	351.67	371.26
5,000,000		514.60	411.67	355.96	351.58	347.85	351.38	368.36

		Line						
		Manufacturing Overhead plus Materials Cost per unit						
K ↓ vs L_daily →		500	1,000	2,500	3,000	4,000	8,000	16,000
100,000		na	na	na	na	na	na	na
200,000		na	na	na	na	na	na	na
300,000		na	na	na	na	na	na	na
400,000		na	na	na	na	na	na	na
500,000		150.64	157.77	175.43	180.46	189.79	221.98	276.86
600,000		149.59	155.91	172.62	177.46	186.48	217.91	271.93
700,000		148.92	154.52	170.41	175.10	183.88	214.67	267.99
800,000		148.49	153.46	168.64	173.19	181.75	212.02	264.74
900,000		148.25	152.63	167.18	171.61	179.99	209.79	262.01
1,000,000		148.14	151.97	165.95	170.28	178.49	207.89	259.67
2,000,000		150.40	150.03	159.92	163.53	170.64	197.45	246.52
3,000,000		154.66	151.13	158.19	161.34	167.78	193.07	240.67
4,000,000		159.40	153.16	157.92	160.72	166.64	190.81	237.38
5,000,000		164.27	155.60	158.35	160.84	166.33	189.60	235.37

		Cell						
		Manufacturing Overhead plus Materials Cost per unit						
K ↓ vs L_daily →		500	1,000	2,500	3,000	4,000	8,000	16,000
100,000		na	na	na	na	na	na	na
200,000		na	na	na	na	na	na	na
300,000		na	na	na	na	na	na	na
400,000		na	na	na	na	na	na	na
500,000		na	na	na	na	na	na	na
600,000		na	na	na	na	na	na	na
700,000		na	na	na	na	na	na	na
800,000		na	na	na	na	na	na	na
900,000		na	na	na	na	na	na	na
1,000,000		na	na	na	na	na	na	na
2,000,000		na	na	na	na	na	na	na
3,000,000		113.82	119.48	136.45	142.06	153.21	197.27	284.01
4,000,000		115.43	120.86	137.34	142.81	153.70	196.88	282.20
5,000,000		117.14	122.43	138.59	143.97	154.70	197.32	281.74

Build New Factory

Globex Region

As of day 368.31

▼ 1 Inventory ↻

10 records Search:

Product	DC	\$	Units	Last date
Name		Price	On hand	Any
Heart View	Globex	700	207	367.94

Showing 1 to 1 of 1 entries

▼ 1 Factory ↻

10 records Search:

Add New +

Team	Name	Region	Status	WIP Units	Tech	Cap EX (\$)	Daily EX (\$)
				Total		Total	Total
Globex	Pilot Factory in Globex	Globex	PROCESSING	100		100,000	2,500

Showing 1 to 1 of 1 entries

▼ 1 Distribution Center ↻

10 records Search:

Add New +

Team	Name	Region	Status
			Current
Globex	Local DC in Globex	Globex	READY

Showing 1 to 1 of 1 entries

Start New Factory

Name

Globex

Factory 1

Land Cost

\$

100,000.00

Capital Expenditure

\$

Technology

===Select===

Build Duration

90

days

Daily Expenditure

\$

SA Clone Factory List

None

CANCEL

CREATE

Confirm Transaction

Factory Transaction Detail

Factory Name : Globex-New Factory

Land Cost : \$100,000.00

Capital Exp : \$10,000.00

Technology : Bench

Build Duration : 90

Daily Exp : \$0.00

When :

Now

Confirm

Cancel

Increase Capital

Pilot Factory in Globex [Globex] [PROCESSING]

Name : Pilot Factory in Globex

Owner : Globex

Location : Globex

Status : PROCESSING

Technology : Bench

Land expenditure : \$ 100,000

Daily expenditure : \$ 2,500

Capital expenditure : \$ 100,000

Operating income : \$ -1,081,703

Inventory in process : 100 Units
of Globex-Heart View
for Local DC in Globex

Inventory in transit : 100 Units
of Globex-Heart View
for Local DC in Globex

Throughput : 9.04 units per day

Overhead per Unit : \$ 278.45

As of day 366.14

1 Outgoing Shipping Agreement

10 records

Search:

Id	Destination	Product	Priority	Terms	Days delayed	Batches	Projected contribution per unit Margin	Status
		Name						
1		Heart View	0	100 @ \$100	0	33	\$ -50	ACTIVE

Showing 1 to 1 of 1 entries

Close

Capital expenditure : \$ 100,000

\$ ✓ ✗

Confirm Transaction

Pending Capital Expenditure will be effective on day
411.00

Factory Name : Pilot Factory in Arryn
Current Cap Exp : 100,000
Requested Cap Exp : 1,000.00
Effective Day : 411.00
When : Now

Confirm

Cancel

Change Labor

Pilot Factory in Globex [Globex] [PROCESSING]

Name : Pilot Factory in Globex

Owner : Globex

Location : Globex

Status : PROCESSING

Technology : Bench

Land expenditure : \$ 100,000

Daily expenditure : **\$ 2,500**

Capital expenditure : \$ 100,000

Operating income : \$ -1,081,703

Inventory in process : 100 Units

Inventory in transit : 100 Units

Throughput : 9.04 units per day

Overhead per Unit : \$ 278.45

As of day 366.14

1 Outgoing Shipping Agreement

10 records

Search:

Id	Destination	Product	Priority	Terms	Days delayed	Batches	Projected contribution per unit	Status
		Name					Margin	
1		Heart View	0	100 @ \$100	0	33	\$ -50	ACTIVE

Showing 1 to 1 of 1 entries

Close

Daily expenditure : \$ 2,500

\$ 0

Confirm Transaction

Factory Change Daily Exp Transaction

Factory Name : Pilot Factory in Arrakis

Current Daily Exp : 2,500.00

Requested Daily Exp : 2,000.00

When : Now

Confirm

Cancel

Shut Down Factory

Pilot Factory in Globex [Globex] [PROCESSING]

Shutdown is pending completion of current batch.

As of day 439.19

Name : Pilot Factory in Globex

Owner : Globex

Location : Globex

Status : PROCESSING

Technology : Bench

Land expenditure : \$ 100,000

Daily expenditure : \$ 2,500

Capital expenditure : \$ 100,000

Operating income : \$ -1,295,540

Inventory in process : 100 Units
of Globex-Heart View
for Local DC in Globex

Inventory in transit : 100 Units
of Globex-Heart View
for Local DC in Globex

Throughput : 9.04 units per day

Overhead per Unit : \$ 278.45

1 Outgoing Shipping Agreement

10 records

Search:

Id	Destination	Product	Priority	Terms	Days delayed	Batches	Projected contribution per unit	Status
		Name					Margin	
1		Heart View	0	100 @ \$100	0	39	\$ -50	ACTIVE

Showing 1 to 1 of 1 entries

Close

Build New Distribution Center

Globex Region

As of day 368.31

1 Inventory

10 records

Search:

Product	DC	\$	Units	Last date																
Name		Price	On hand	Any																
Heart View		700	207	367.94																

Showing 1 to 1 of 1 entries

1 Factory

10 records

Search:

	Team	Name	Region	Status	WIP Units	Tech	Cap EX (\$)	Daily EX (\$)
							Total	Total
	Globex	Pilot Factory in Globex	Globex	PROCESSING	100		100,000	2,500

Showing 1 to 1 of 1 entries

1 Distribution Center

10 records

Search:

	Team	Name	Region	Status
				Current
	Globex	Local DC in Globex	Globex	READY

Showing 1 to 1 of 1 entries

Start New Distribution Center

Name

Globex- DC 1

Region

Globex

Land Cost

\$100,000

Capital Expenditure

\$2,500,000

Build Duration

60 days

Daily Expenditure

\$2,000

Fulfillment Cost

\$10 + 20 % of retail price

SA Clone Building List

None

CANCEL

CREATE

Confirm Transaction

DC Transaction Detail

DC Name : Globex-DC 1

Land Cost : \$100,000

Capital Expenditure : \$2,500,000

Build Duration : 60 days

Daily Expenditure : \$2,000

When : Now

Confirm

Cancel

Shut Down Distribution Center

Local DC in Globex [Globex] [READY]

Name : Local DC in Globex
Owner : Globex
Location : Globex
Status : READY ✎
Operating income : \$ 2,564,186 📊

Total advertising : \$ 44,062.84 📊
Total inventory : 50 units 📊
On-hand inventory : 0 units 📊
Price range : \$ 600 to \$ 1,000 📊
Total retail sales : 11,900 units 📊

Name : Local DC in Globex
Owner : Globex
Location : Globex
Status : READY ✎

Total advertising : \$ 44,062.84 📊
Total inventory : 50 units 📊
On-hand inventory : 0 units 📊
Price range : \$ 600 to \$ 1,000 📊
Total retail sales : 11,900 units 📊

Pending status : SHUTDOWN ✔ ✖

Operating income : \$ 2,564,186 📊

Confirm Transaction

Stop accepting orders now and shut down DC once
there is no remaining inventory?

When :

Confirm

Cancel

Confirm Transaction

Distribution Center ShutDown is
Scheduled Successfully.

Close

Local DC in Globex [Globex] [READY]

Shutdown is pending. Not placing new orders.

As of day 919.87 🔄

Name : Local DC in Globex
Owner : Globex
Location : Globex
Status : READY ✎
Operating income : \$ 2,558,773 📊

Total advertising : \$ 44,081.55 📊
Total inventory : 75 units 📊
On-hand inventory : 25 units 📊
Price range : \$ 600 to \$ 1,000 📊
Total retail sales : 11,925 units 📊

Distribution

- Direct from factory to DC
- Factory to central DC
 - Hub and Spoke
- Priorities

Marketing

Bass Model, Imitators, Innovators, Advertising

Marketing

All regions have the same size markets, with the following exceptions:

- Metropolis has larger markets – up to 2x other regions' markets.
- Serenity is the only region with military markets.
- Serenity has very small non-military markets.

Customer arrival rates determined by the Bass model and advertising spend

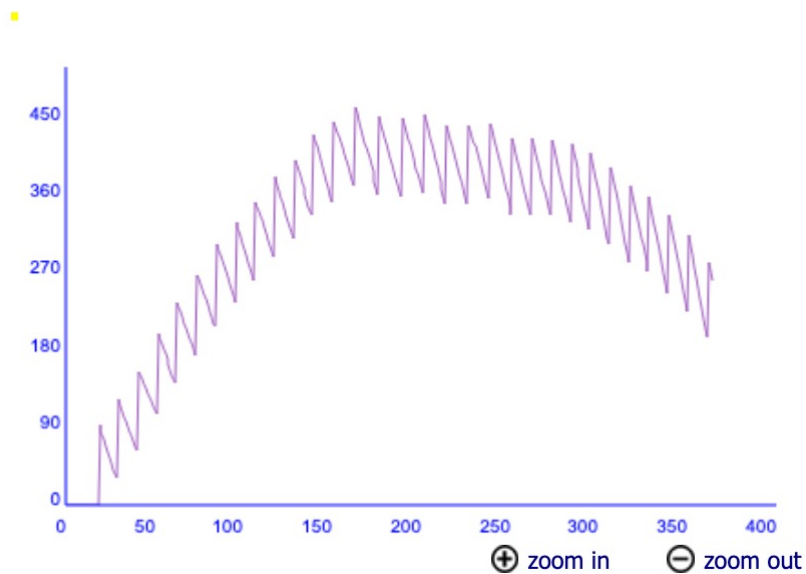
- Innovators (p parameter)
- Imitators (q parameter)
- Advertising (\$500 per additional p)

Distribution

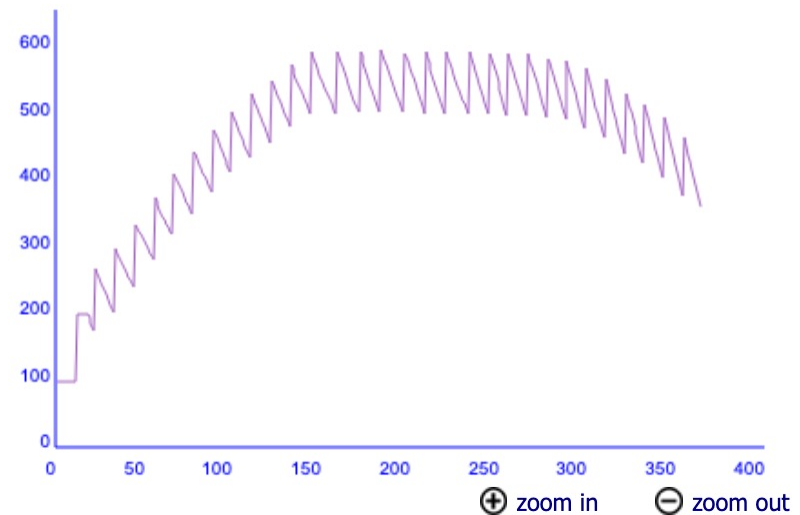
- Factory to DC
- DC to DC
- Retail customers only buy from DCs

Inventory Balances in Year 1

End-of-day On-hand Inventory at Local DC in Westeros



End-of-day Total Inventory at Local DC in Westeros



Daily Sales in Year 1

Name : Local DC in Westeros

Owner : Westeros

Location : Westeros

Status : READY

Operating income : \$ 381,983 

Total advertising : \$ 0 

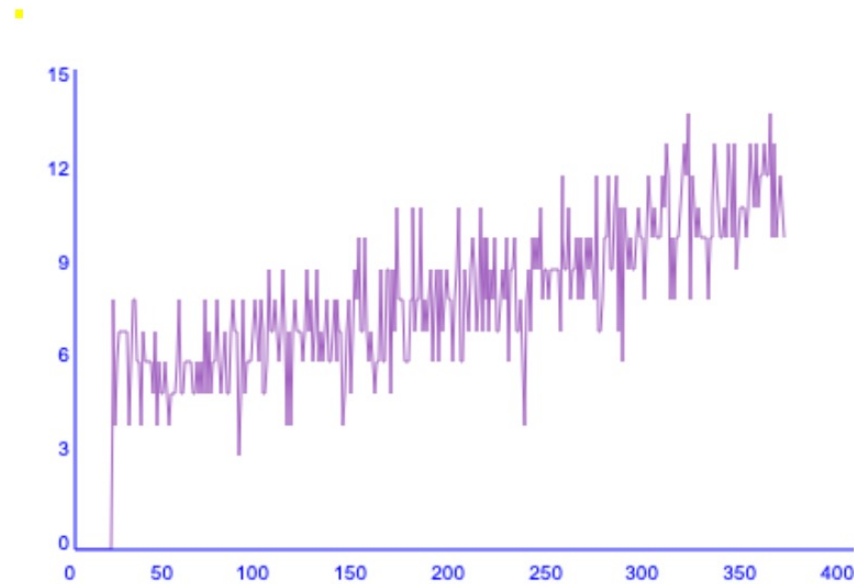
Total inventory : 363 units 

On-hand inventory : 263 units 

Price : \$ 700 

Total retail sales : 2,837 units 

Daily Retail Sales from Local DC in Westeros



Estimating Demand – Bass Model

Who is coming to market and when will they buy?

- Innovators
 - Price shop (in the region)
 - Purchase based on price
 - Purchase the product that gives them the greatest WTP - Price
 - Daily Arrivals = $p * \text{remaining market size}$
- Imitators
 - Only come to your store
 - If $\text{WTP} > \text{Price}$ then they buy
 - Number of imitators coming to your store depends on how much of the market you have served
 - Daily Arrivals = $q * \text{remaining market size} * \% \text{ of the market served}$
- Advertising
 - You can pay to bring people to your store
 - If $\text{WTP} > \text{Price}$ then they buy
 - Daily Arrivals = $p * \text{remaining market size} * \text{advertising spend} / \500
 - We recommend you do not advertise today as you don't really have the capacity to keep up with it. Wait until next week when you have a faster factory.

Bass Model for Arrival Rates

Assumptions:

Innovators, $p = 0.0002$

Imitators, $q = 0.0035$

Advertising = \$0

Market size = 34,500

Bass Model for Arrival Rates

Assumptions:

Innovators, $p = 0.0002$

Imitators, $q = 0.0035$

Advertising = \$0

Market size = 34,500

Innovator arrivals = $p * \text{remaining market size}$

Imitator arrivals = $q * \text{remaining market size} * \% \text{ of the market served}$

Day 1:

Innovators arrivals = $0.0002 * 34,500 = 6.90$

Imitator arrivals = $0.0035 * 34,500 * 0 / 34,500 = 0$

Day 2:

Innovators arrivals = $0.0002 * 34,993.10 = 6.8986$

Imitator arrivals = $0.0035 * 34,993.10 * 6.90 / 34,500 = 0.0241$

Bass Model Spreadsheet

No Advertising

Enter inputs only in yellow cells. If you don't enter a period end day, the model calculates up to day 2,930. If you don't enter another type of input for a future period, the model uses the value from the latest period with an input.

Market Name	MD - Heart - Temporal	Initial Market Size	34,500	Historic Retail Units Sold	0	Quarter End Days									
Market research estimate of initial market size in a regular region: 20,000-40,000.				Mean of WTP		\$723	Historic Retail Revenue		\$0	Q1: 91	Q5: 455	Q9: 819	Q13: 1183	Q17: 15	
Market research estimate of initial willingness to pay (WTP) range: \$600-\$865.				Standard Deviation of WTP		\$30	Historic COGS		\$0	Q2: 182	Q6: 546	Q10: 910	Q14: 1274	Q18: 16	
Every \$500 spent on advertising generates arrivals = 0.0002 * the remaining market.				Approx. WTP Range (mean ± 3s)		\$633	\$813	Historic Retail Fulfillment		\$0	Q3: 273	Q7: 637	Q11: 1001	Q15: 1365	Q19: 17
Innovator arrivals = 0.0002 * the remaining market.				Begin Forecast On Game Day #		19	End Forecast on Game Day #		2,930	Q4: 364	Q8: 728	Q12: 1092	Q16: 1456	Q20: 18	
Imitators arrivals = 0.0035 * the remaining market * the share of the initial market that has purchased your product.															

[illegible]

Bass Model Spreadsheet

No Advertising

Daily Calculations

	Adv. (p / \$500)	Innovation (p)	Imitation (q)
Coefficient	0.0002	0.0002	0.0035

Notes: BOP = Beginning of Period; EOP = End of Period

Total sales cannot be greater than either total arrivals or the remaining market size.

Search by Day:	Day (t)	Retail Price per Unit	\$ of Advertising	Advertising Arrivals ($Q_{a,t}$)	Innovator Arrivals ($Q_{p,t}$)	Imitator Arrivals ($Q_{q,t}$)	Total Arrivals (Q_t)	Sales Rate	Advertising Sales ($A_{a,t}$)	Innovator Sales ($A_{p,t}$)	Imitator Sales ($A_{q,t}$)	Total Sales (A_t)	BOP Cumulative Sales (A)	BOP Market Penetration % (A/M)	BOP Remaining Market Size (M-A)	EOP Cumulative Sales (A+A _t)	EOP Market Penetration % ($[A+A_t]/M$)
	19	\$ 700.00	\$ -	-	6.90	-	6.90	77.94%	-	5.38	-	5.38	-	0.00%	34,500	5	0.02%
	364	\$ 700.00	\$ -	-	6.33	9.18	15.51	75.94%	-	4.81	6.97	11.78	2,861	8.29%	31,639	2,872	8.33%
	728	\$ 700.00	\$ -	-	5.17	22.70	27.87	70.53%	-	3.65	16.01	19.66	8,661	25.10%	25,839	8,681	25.16%
	1,092	\$ 700.00	\$ -	-	3.67	30.07	33.74	58.43%	-	2.14	17.57	19.71	16,171	46.87%	18,329	16,190	46.93%

Quarter	Day (t)	Retail Price per Unit	\$ of Advertising	Advertising Arrivals ($Q_{a,t}$)	Innovator Arrivals ($Q_{p,t}$)	Imitator Arrivals ($Q_{q,t}$)	Total Arrivals (Q_t)	Sales Rate	Advertising Sales ($A_{a,t}$)	Innovator Sales ($A_{p,t}$)	Imitator Sales ($A_{q,t}$)	Total Sales (A_t)	BOP Cumulative Sales (A)	BOP Market Penetration % (A/M)	BOP Remaining Market Size (M-A)	EOP Cumulative Sales (A+A _t)	EOP Market Penetration % ($[A+A_t]/M$)
1	19	\$ 700.00	\$ -	-	6.90	-	6.90	77.94%	-	5.38	-	5.38	-	0.00%	34,500	5	0.02%
1	20	\$ 700.00	\$ -	-	6.90	0.02	6.92	77.94%	-	5.38	0.01	5.39	5	0.02%	34,495	11	0.03%
1	21	\$ 700.00	\$ -	-	6.90	0.04	6.94	77.93%	-	5.38	0.03	5.41	11	0.03%	34,489	16	0.05%
1	22	\$ 700.00	\$ -	-	6.90	0.06	6.95	77.93%	-	5.37	0.04	5.42	16	0.05%	34,484	22	0.06%
1	23	\$ 700.00	\$ -	-	6.90	0.08	6.97	77.93%	-	5.37	0.06	5.43	22	0.06%	34,478	27	0.08%
1	24	\$ 700.00	\$ -	-	6.89	0.09	6.99	77.92%	-	5.37	0.07	5.45	27	0.08%	34,473	32	0.09%
1	25	\$ 700.00	\$ -	-	6.89	0.11	7.01	77.92%	-	5.37	0.09	5.46	32	0.09%	34,468	38	0.11%
1	26	\$ 700.00	\$ -	-	6.89	0.13	7.03	77.92%	-	5.37	0.10	5.47	38	0.11%	34,462	43	0.13%
1	27	\$ 700.00	\$ -	-	6.89	0.15	7.04	77.91%	-	5.37	0.12	5.49	43	0.13%	34,457	49	0.14%
1	28	\$ 700.00	\$ -	-	6.89	0.17	7.06	77.91%	-	5.37	0.13	5.50	49	0.14%	34,451	54	0.16%
1	29	\$ 700.00	\$ -	-	6.89	0.19	7.08	77.91%	-	5.37	0.15	5.52	54	0.16%	34,446	60	0.17%
1	30	\$ 700.00	\$ -	-	6.89	0.21	7.10	77.90%	-	5.37	0.16	5.53	60	0.17%	34,440	65	0.19%
1	31	\$ 700.00	\$ -	-	6.89	0.23	7.12	77.90%	-	5.36	0.18	5.54	65	0.19%	34,435	71	0.21%
1	32	\$ 700.00	\$ -	-	6.89	0.25	7.13	77.90%	-	5.36	0.19	5.56	71	0.21%	34,429	77	0.22%
1	33	\$ 700.00	\$ -	-	6.88	0.27	7.15	77.89%	-	5.36	0.21	5.57	77	0.22%	34,423	82	0.24%
1	34	\$ 700.00	\$ -	-	6.88	0.29	7.17	77.89%	-	5.36	0.22	5.58	82	0.24%	34,418	88	0.25%
1	35	\$ 700.00	\$ -	-	6.88	0.31	7.19	77.88%	-	5.36	0.24	5.60	88	0.25%	34,412	93	0.27%
1	36	\$ 700.00	\$ -	-	6.88	0.33	7.21	77.88%	-	5.36	0.25	5.61	93	0.27%	34,407	99	0.29%
1	37	\$ 700.00	\$ -	-	6.88	0.35	7.23	77.88%	-	5.36	0.27	5.63	99	0.29%	34,401	105	0.30%
1	38	\$ 700.00	\$ -	-	6.88	0.36	7.24	77.87%	-	5.36	0.28	5.64	105	0.30%	34,395	110	0.32%

Bass Model Spreadsheet

\$500 Advertising

Enter inputs only in yellow cells. If you don't enter a period end day, the model calculates up to day 2,930. If you don't enter another type of input for a future period, the model uses the value from the latest period with an input.

Market Name	MD - Heart - Temporal																															
Market research estimate of initial market size in a regular region: 20,000-40,000.																																
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Begin Forecast On Game Day #	19																															
Historic Retail Units Sold	0																															
Historic Retail Revenue	\$0																															
Historic COGS	\$0																															
Historic Retail Fulfillment	\$0																															
End Forecast on Game Day #	2,930																															
<table><tr><th colspan="6">Quarter End Days</th></tr><tr><td>Q1: 91</td><td>Q5: 455</td><td>Q9: 819</td><td>Q13: 1183</td><td>Q17: 15</td><td></td></tr><tr><td>Q2: 182</td><td>Q6: 546</td><td>Q10: 910</td><td>Q14: 1274</td><td>Q18: 16</td><td></td></tr><tr><td>Q3: 273</td><td>Q7: 637</td><td>Q11: 1001</td><td>Q15: 1365</td><td>Q19: 17</td><td></td></tr><tr><td>Q4: 364</td><td>Q8: 728</td><td>Q12: 1092</td><td>Q16: 1456</td><td>Q20: 18</td><td></td></tr></table>			Quarter End Days						Q1: 91	Q5: 455	Q9: 819	Q13: 1183	Q17: 15		Q2: 182	Q6: 546	Q10: 910	Q14: 1274	Q18: 16		Q3: 273	Q7: 637	Q11: 1001	Q15: 1365	Q19: 17		Q4: 364	Q8: 728	Q12: 1092	Q16: 1456	Q20: 18	
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[illegible]

Bass Model Spreadsheet

\$500 Advertising

Daily Calculations

	Adv. (p / \$500)	Innovation (p)	Imitation (q)
Coefficient	0.0002	0.0002	0.0035

Notes: BOP = Beginning of Period; EOP = End of Period

Total sales cannot be greater than either total arrivals or the remaining market size.

Search by Day:	Day (t)	Retail Price per Unit	\$ of Advertising	Advertising Arrivals ($Q_{a,t}$)	Innovator Arrivals ($Q_{p,t}$)	Imitator Arrivals ($Q_{q,t}$)	Total Arrivals (Q_t)	Sales Rate	Advertising Sales ($A_{a,t}$)	Innovator Sales ($A_{p,t}$)	Imitator Sales ($A_{q,t}$)	Total Sales (A_t)	BOP Cumulative Sales (A)	BOP Market Penetration % (A/M)	BOP Remaining Market Size (M-A)	EOP Cumulative Sales (A+A _t)	EOP Market Penetration % ($[A+A_t]/M$)
	19	\$ 700.00	\$ 500.00	6.90	6.90	-	13.80	77.94%	5.38	5.38	-	10.76	-	0.00%	34,500	11	0.03%
	364	\$ 700.00	\$ 500.00	5.83	5.83	15.80	27.47	73.89%	4.31	4.31	11.68	20.30	5,343	15.49%	29,157	5,363	15.55%
	728	\$ 700.00	\$ 500.00	4.15	4.15	28.94	37.24	63.31%	2.63	2.63	18.32	23.58	13,741	39.83%	20,759	13,765	39.90%
	1,092	\$ 700.00	\$ 500.00	2.71	2.71	28.79	34.21	43.69%	1.18	1.18	12.58	14.95	20,956	60.74%	13,544	20,971	60.78%

Quarter	Day (t)	Retail Price per Unit	\$ of Advertising	Advertising Arrivals ($Q_{a,t}$)	Innovator Arrivals ($Q_{p,t}$)	Imitator Arrivals ($Q_{q,t}$)	Total Arrivals (Q_t)	Sales Rate	Advertising Sales ($A_{a,t}$)	Innovator Sales ($A_{p,t}$)	Imitator Sales ($A_{q,t}$)	Total Sales (A_t)	BOP Cumulative Sales (A)	BOP Market Penetration % (A/M)	BOP Remaining Market Size (M-A)	EOP Cumulative Sales (A+A _t)	EOP Market Penetration % ($[A+A_t]/M$)
1	19	\$ 700.00	\$ 500.00	6.90	6.90	-	13.80	77.94%	5.38	5.38	-	10.76	-	0.00%	34,500	11	0.03%
1	20	\$ 700.00	\$ 500.00	6.90	6.90	0.04	13.83	77.93%	5.38	5.38	0.03	10.78	11	0.03%	34,489	22	0.06%
1	21	\$ 700.00	\$ 500.00	6.90	6.90	0.08	13.87	77.93%	5.37	5.37	0.06	10.81	22	0.06%	34,478	32	0.09%
1	22	\$ 700.00	\$ 500.00	6.89	6.89	0.11	13.90	77.92%	5.37	5.37	0.09	10.83	32	0.09%	34,468	43	0.13%
1	23	\$ 700.00	\$ 500.00	6.89	6.89	0.15	13.93	77.91%	5.37	5.37	0.12	10.86	43	0.13%	34,457	54	0.16%
1	24	\$ 700.00	\$ 500.00	6.89	6.89	0.19	13.97	77.91%	5.37	5.37	0.15	10.88	54	0.16%	34,446	65	0.19%
1	25	\$ 700.00	\$ 500.00	6.89	6.89	0.23	14.00	77.90%	5.36	5.36	0.18	10.91	65	0.19%	34,435	76	0.22%
1	26	\$ 700.00	\$ 500.00	6.88	6.88	0.26	14.03	77.89%	5.36	5.36	0.21	10.93	76	0.22%	34,424	87	0.25%
1	27	\$ 700.00	\$ 500.00	6.88	6.88	0.30	14.07	77.89%	5.36	5.36	0.24	10.96	87	0.25%	34,413	98	0.28%
1	28	\$ 700.00	\$ 500.00	6.88	6.88	0.34	14.10	77.88%	5.36	5.36	0.27	10.98	98	0.28%	34,402	109	0.32%
1	29	\$ 700.00	\$ 500.00	6.88	6.88	0.38	14.14	77.87%	5.36	5.36	0.30	11.01	109	0.32%	34,391	120	0.35%
1	30	\$ 700.00	\$ 500.00	6.88	6.88	0.42	14.17	77.86%	5.35	5.35	0.33	11.03	120	0.35%	34,380	131	0.38%
1	31	\$ 700.00	\$ 500.00	6.87	6.87	0.46	14.20	77.86%	5.35	5.35	0.35	11.06	131	0.38%	34,369	142	0.41%
1	32	\$ 700.00	\$ 500.00	6.87	6.87	0.49	14.24	77.85%	5.35	5.35	0.38	11.08	142	0.41%	34,358	153	0.44%
1	33	\$ 700.00	\$ 500.00	6.87	6.87	0.53	14.27	77.84%	5.35	5.35	0.41	11.11	153	0.44%	34,347	164	0.48%
1	34	\$ 700.00	\$ 500.00	6.87	6.87	0.57	14.31	77.84%	5.35	5.35	0.44	11.13	164	0.48%	34,336	175	0.51%
1	35	\$ 700.00	\$ 500.00	6.86	6.86	0.61	14.34	77.83%	5.34	5.34	0.47	11.16	175	0.51%	34,325	186	0.54%
1	36	\$ 700.00	\$ 500.00	6.86	6.86	0.65	14.37	77.82%	5.34	5.34	0.50	11.19	186	0.54%	34,314	197	0.57%
1	37	\$ 700.00	\$ 500.00	6.86	6.86	0.69	14.41	77.81%	5.34	5.34	0.53	11.21	197	0.57%	34,303	209	0.60%
1	38	\$ 700.00	\$ 500.00	6.86	6.86	0.73	14.44	77.81%	5.34	5.34	0.56	11.24	209	0.60%	34,291	220	0.64%

Back to Basics

SUPPLY = DEMAND

Financing

How will we finance our investing and operating activities?

- Retained Earnings
- Working Capital
- Borrow
 - 5-year zero coupon bonds
 - Excellent, 10% APR, 20x interest coverage ratio
 - Good, 15% APR, 7x interest coverage ratio
 - Poor, 25% APR, 2x interest coverage ratio
 - Daily compounding

When can we borrow?

- Need a full quarter of positive EBIT (operating income)
- Borrowing based on your interest coverage ratio
- Interest coverage ratio = $(4 * \text{quarterly EBIT}) / (\text{interest expense if you borrow})$
- 5-year zero coupon bonds
- Daily compounding
- Effective annual rate = $(1 + \text{APR}/n)^n - 1$

Issuing Bonds Example

EBIT Q5 = \$25,000

How much can we borrow at excellent?

Issuing Bonds Example

EBIT Q5 = \$25,000

How much can we borrow at good?

Issuing Bonds Example

EBIT Q5 = \$25,000

How much can we borrow at poor?

Debt Model

\$25,000 in Quarterly EBIT

Inputs

What is your current quarterly interest expense?	\$0	<i>per quarter</i>	\$0	<i>per year</i>
What was your operating income for the last full quarter?	\$25,000	<i>per quarter</i>	\$100,000	<i>per year</i>

Credit Rating Criteria and Interest Rates

	Excellent	Good	Poor
Interest Coverage Hurdle for Credit Rating	20x	7x	2x
Annual Percentage Rate (APR)	10.00%	15.00%	25.00%
Effective Annual Rate (EAR)	10.516%	16.180%	28.392%

Maximum Interest Expense to Retain Credit Rating

	Excellent	Good	Poor	Total
Cumulative Annualized Interest Expense	\$5,000.00	\$14,285.71	\$50,000.00	\$50,000.00
Incremental Annualized Interest Expense	\$5,000.00	\$9,285.71	\$35,714.29	\$50,000.00

Debt Capacity

	Excellent	Good	Poor	Total
# of Issuable Bonds	78	121	439	639
Face Value of Issuable Bonds	\$78,388.88	\$121,477.27	\$438,869.13	\$638,735.27
Present Value of Issuable Bonds (Cash Received)	\$47,548.52	\$57,390.66	\$125,792.07	\$230,731.26

Value of a Single Bond

	Excellent	Good	Poor
Face Value of One Bond	\$1,000.00	\$1,000.00	\$1,000.00
Present Value of One Bond	\$606.57	\$472.44	\$286.63

Issuing Bonds Example Solutions

EBIT Q5 = \$25,000

How much interest expense can we support at the Excellent rate?	$\$100,000 / \text{interest expense} = 20$ interest expense = \$5,000
What is the present value of the debt that would generate the interest expense above?	$r = (1 + 0.10 / 364)^{364} - 1 = 0.10516$ $\$5,000 = PV * 0.10516$ PV = \$47,548.52
What is the future value (face value) of debt that corresponds to the present value of the debt calculated above?	$\$47,548.52 = FV / (1 + 0.10516)^5$ FV = \$79,388.88
How much interest expense can we support at the Good rate?	$\$100,000 / 7 = \$14,285.71$
Subtract the interest expense already allocated to the Excellent debt. This is how much interest expense can be allocated to the Good debt.	$\$14,285.71 - \$5,000 = \$9,285.71$
What is the present value of the debt that would generate the interest expense above?	$r = (1 + 0.15 / 364)^{364} - 1 = 0.1618$ $\$9,285.71 = PV * 0.1618$ PV = \$57,390.66
What is the future value (face value) of debt that corresponds to the present value of the debt calculated above?	$\$57,390.66 = FV / (1.1618)^5$ FV = \$121,477.27
How much interest expense can we support at the Poor rate?	$\$100,000 / 2 = \$50,000$
Subtract the interest expense already allocated to the Excellent and Good debt. This is how much interest expense can be allocated to the Poor debt.	$\$50,000 - \$14,285.71 = \$35,714.29$
What is the present value of the debt that would generate the interest expense above?	$r = (1 + 0.25 / 364)^{364} - 1 = 0.283915$ $\$35,714.29 = PV * 0.283915$ PV = \$125,792.07
What is the future value (face value) of debt that corresponds to the present value of the debt calculated above?	$\$125,792.07 = FV / (1.283915)^5$ FV = \$438,869.13

Borrowing

Headquarters – you can't borrow when you start as you have negative EBIT

BarterTown Headquarters



Hover over values for details. Click on numbers and icons to open panels and edit values.

As of day 364 ↺

Finance

Operating Income : \$-105,160 (Q 4)

Cash : \$1,579,530

Working capital ratio : 18.08

Trade credit : materials net / 30
others net / 15

Bonds : \$0 0 issuable

Dividends : \$0

Marketing

Designs : 0 ... 0  1  0 

Retail sales : 1 of 102 regional markets
1 inventories
\$0 per day advertising
2,837 units sold total
\$1,985,900 total revenues

Sales to other teams : 0 inventories
0 units sold total
\$0 total revenues

Operations

Facilities :  

All agreements : 0 ... 1  0 

263 units on hand
0 units in transit

Production : 0 ... 1  0 

100 units in process

Procurement : 0 ... 0  0 

Fulfillment : 0 ... 0  0 

Long-Term Debt Panel for Bonds

Long-term debt

Up to 1,680 bonds issuable at Excellent (10% YTM)

0 Bond Issue     **ADD NEW +**

10 records Search:

Day	Terms	Count			Value	YTM	Days to maturity	Redemption price
		Total	Outstanding	Redeemed				
					Face			

No data available in table

Long-term debt

Up to 2,584 bonds issuable at Good (15% YTM)

1 Bond Issue     **ADD NEW +**

10 records Search:

Day	Terms	Count			Value	YTM	Days to maturity	Redemption price
		Total	Outstanding	Redeemed				
					Face			
838	5 zero cpn 	1,680	1,680	0 	\$1,680,000	10%	1,819	\$614.07

Showing 1 to 1 of 1 entries  **1** 

Bond Terms Panel : BarterTown

As of day 830.12 

Rating : Excellent
Issue price : \$ 613.91
Maturity : 5years 
Issue count : 1680 
Par : \$1,000
Yield to maturity : 10% per annum
Coupon rate : 0% 
 Callable at accreted value starting immediately

Notes

Practice Game

Important Links and Documents

See Class 3 on Canvas

- Practice Game Link
- Important Documents
 - Quick Reference for Practice Game
 - Market Research for Practice Game
 - Gleacher Game Case Brief
 - Product Design Guide
 - Suggested Team Roles
 - Player Manual
- Models (all complete, with formulas, adapt to your needs)
 - Production Model
 - Bass Model
 - Debt Model

Chat is in Envelopes

Envelope turns yellow when there's a new message in a closed chat window

- The group chat icon  is to the right of game status above the map.
 - Use this to contact all teams during a game.
- The admin chat icon  is to the right of group chat.
 - Use this to contact just Kathleen or the TAs during a game.
- If you want to send a message to a specific team, click on the envelope  in their home region on the map.
- If there is a new message, the chat icon turns yellow.
 - The icon will not turn yellow if the chat window is already open.



Practice Game

Play

- Start at the end of quarter 4 (day 365) and end on day 1092
- Everyone starts at the same point, which is where we started yesterday.
- Everything you learned yesterday applies today.
- Play 3 quarters in class, from 4:45pm-6:15pm
- Play 3 more quarters tonight, from 7pm-8:30pm, ending live play on day 910.
- Speed: 2 quarters per hour, 30 minutes per quarter, 3 game days per minute.
- We will run the game quickly through day 1092, so you can see what would have happened if your company ran two more quarters.
- **Please do the homework together – it is important to all be on the same page with your strategy. The homework will take approximately 1.5-2 hours. It is the last long assignment this week.**

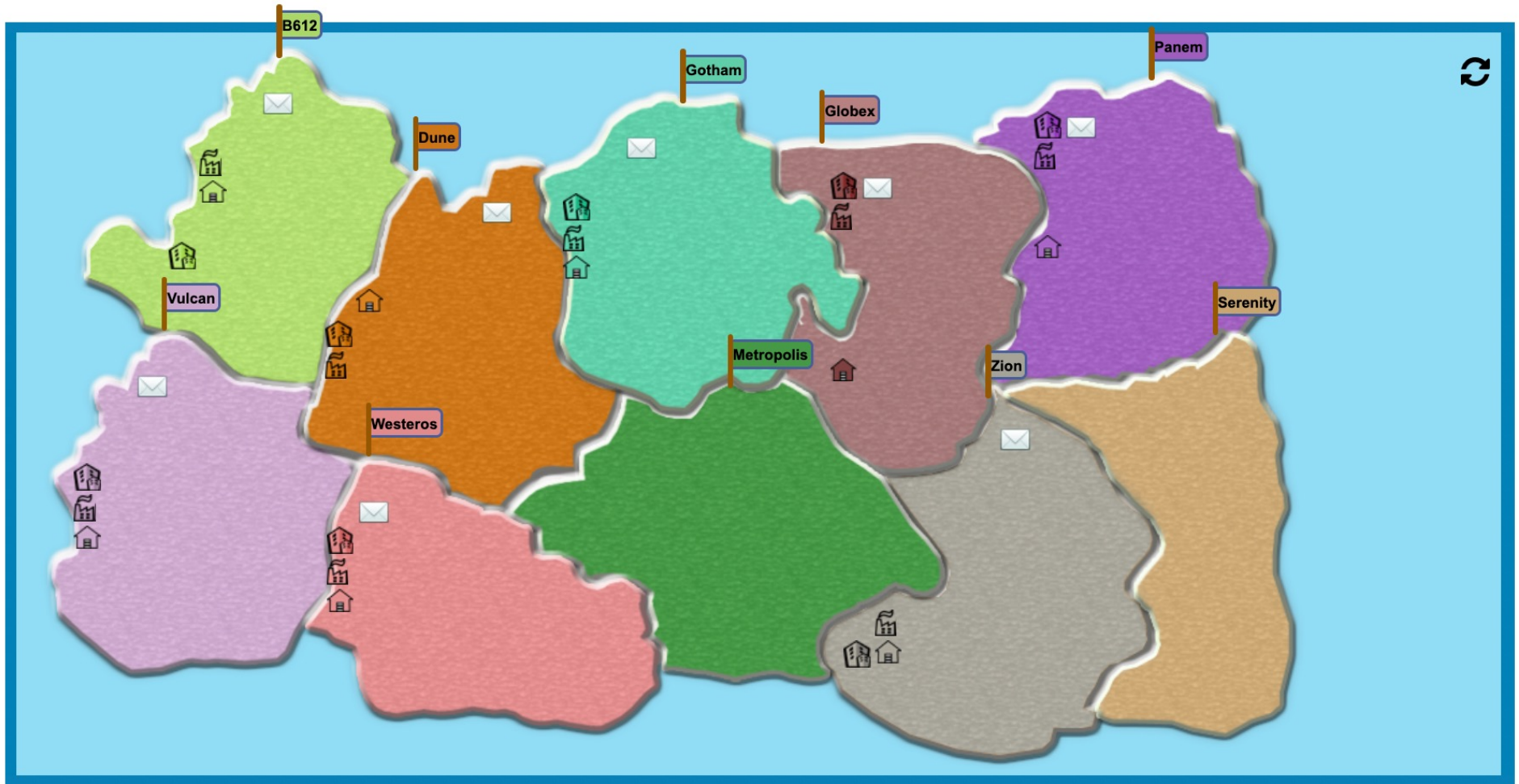
Things to Consider

Use the Quick Reference
Guide!

- Market sizes are the same in all regions, with the exception of Metropolis
- Willingness to pay is the same in all regions
- You can trade with all teams
- You can change your batch sizes
- You can speed up or slow down your factory
 - Through CAPEX
 - Through Daily Expenditure
- You can use containers or mail for shipping
 - Consider time
 - Consider how many units you need in a container to equal the per unit mailing cost
- You can ship to wholesale customers from the factory or from your DC
- You can only sell to retail customers through the DCs
- Retailers pay handling and commission

Practice Game Map

Serenity (military) and Metropolis (large) have no home teams



Starting Point

World	Team Name	Return to Investors	Retained Earnings at end of day 364	Cash Position
	B612	\$ <u>1,579,530</u>	\$ -342,398	\$ <u>1,579,530</u>
	Dune	\$ <u>1,579,530</u>	\$ -342,398	\$ <u>1,579,530</u>
	Globex	\$ <u>1,579,530</u>	\$ -342,398	\$ <u>1,579,530</u>
	Gotham	\$ <u>1,579,530</u>	\$ -342,398	\$ <u>1,579,530</u>
	Panem	\$ <u>1,579,530</u>	\$ -342,398	\$ <u>1,579,530</u>
	Vulcan	\$ <u>1,579,530</u>	\$ -342,398	\$ <u>1,579,530</u>
	Westeros	\$ <u>1,579,530</u>	\$ -342,398	\$ <u>1,579,530</u>
	Zion	\$ <u>1,579,530</u>	\$ -342,398	\$ <u>1,579,530</u>

Practice Game – Big Picture

- Start with one product
 - Heart View
- Build a new factory to help become profitable
 - You have \$1,549,077 now – extra cash for practice
 - Use Production Model
- Design, build and sell products
 - Advertise
 - Use Bass Model
- Borrow money
 - Use Debt Model
- Expand your business into new regions
 - Borrow and build a DC
 - Make partnerships

Suggested Steps

- Build a new line factory and **give it a name**
 - Schedule the old one to close shortly after the new one opens
 - Clone your shipping agreements
- Build a new product or two and **give them names**
 - Make sure they can be profitable in your line factory
 - Design, develop, price
 - Set up a shipping agreement with your new factory to produce it
 - Set up shipping agreements with competitors to sell it
 - Don't forget to make your re-order points > 0
- Focus on running a profitable business
 - As soon as you have a profitable quarter, you can borrow cash
- Borrow and expand
 - Products
 - Distribution Center
 - Factories

Markets for Today

See the Market Research Report for Practice Game for details

- Athlete
- MD Heart
- MD Dissolved Gasses
- MD Cancer (Breast)
- MD Cancer (Bladder & Kidney)
- MD Fertility (Estrogen)
- MD Metabolic (Bilirubin)
- Law (Narcotic)
- Clinical Fertility
- Clinical Cardiovascular
- Military (Botulinum)
- Military (Anatoxin)

Use the Reports

See the Player Manual for details

- Access reports through the HQ
 - Financial Statements
 - Working Capital Analysis
 - Product Design
 - Product Line Architecture
 - Regional Markets
 - Market Grid
 - Focus Groups
 - Consolidated Inventories
 - Inventories Grid
 - Consolidated Shipping Agreements
- “Open Report” from the map screen
 - Financial Statements
 - Working Capital Analysis
 - Market Grid
 - Product Analysis
 - Price Response
 - Product Line Architecture
 - Product Designs
 - Focus Groups
 - Inventory Status
 - Inventory Grid
 - Active Shipping Agreement Sankey
 - Factories
 - Distribution Centers
 - Active Factories Grid

Things to Remember

Seriously, read these!

- You can build new factories.
- You can build new distribution centers.
- Set up shipping agreements at the same time you start construction on factories and DCs, and at the same you start development of new products.
- Name your factories and products in a way that makes them easy to identify.
- You can speed up or slow down your factory. Never set labor to a very low amount. Set it either to 0 to shut down the factory or to something reasonable for production. Only add incremental capital to a factory.
- Understand why factories go idle and what to do about it. Understand why you stock out and what to do about it. Watch your inventory status report to identify service interruptions and build up of inventory.
- Make sure the products you design or purchase include the features you think they do.
- Make sure you run focus groups for the correct markets.
- Don't sell things for less than what they cost you to produce.
- You can trade with anyone. Agree on terms through chat or in person before sending shipping agreements.
- When sending shipping agreements to other teams, include some WTP information in the comment section to cut down on negotiation time.
- Remember that if you find a market in your region, the same market, WTP and market size exists in all other regions.
- Don't forget that the commission and handling charges are paid by the retailer and shipping is paid by the wholesaler. Incorporate that information in your offers.
- You can use containers or mail for shipping.
- You can ship to wholesale customers from your DC or your factory.
- You can only sell to retail customers through the DCs.
- Make sure you understand how priorities and reorder points work. Make your reorder points > 0 for active shipping agreements. To suspend a shipping agreement, put the re-order point at -1 .

Things to Remember

Seriously, read these!

- Use the scheduling features to shut down factories, start and stop advertising, schedule construction projects, start development, among other things.
- Review your financials frequently to catch problems.
- Make sure the left hand knows what the right hand is doing.
- Choose strategic partners wisely.
- Do not use outdated course materials; only use those for today.
- Use the Quick Reference guide.
- Use the Player Manual.
- Use the reports.
- Use your models. We made them for a reason.
- Watch the chat panel as we put tips and answer questions there, and other teams may be trying to contact you.

For More Tips

CHECK CHAT

Back to Basics

SUPPLY = DEMAND

Objective

MAXIMIZE CURRENT SHAREHOLDER VALUE

(Your company is a going concern – operate for the long run)

* WARNING – REAL GAME STARTING POINT *

- You will only have \$549K in cash when we start the real game on Wednesday
- After changing container to mail – you will need to buy a production line, but you don't have \$600,000.
- You will need a PLAN to get to \$600K.
- You should NOT run focus groups or build any new products until you have the production line under construction.
- Please run numbers and have a concrete plan to get to \$600K. This is part of Assignment 4.
- You have all the data you need, so put your analytical minds to work.

Good Luck!

Appendix: Possible Roles

CEO

- Responsible for overall success and strategy of the business for the foreseeable future.
- Develop a strategy that positions the company to maintain a competitive advantage beyond the duration of the simulation.
- Lead the senior management team.
- Facilitate cooperation and communication among the leaders of various functions or business units.
- Anticipate changes in the competitive landscape by identifying and tracking key performance indicators.
- Work with CMO and CSO to identify when to enter and exit markets.
- Work with the CFO and COO to make capital allocation decisions.
- Work with the CFO to determine your growth and dividend payout strategy.
- Work closely with the leadership team to build the strategy for the period following the simulation and manage the final simulation write-up.

CFO

- Manage cash flows from operating, investing and financing activities.
- Prepare financial projections and budgets for business.
- Manage working capital through trade credit terms, DSO and other tools at your disposal.
- Manage the debt and credit ratings.
- Work with the COO to determine the appropriate transfer price from factory to distribution center.
- Work with the COO to develop appropriate cost measures to support pricing and production decisions.
- Work with the CEO and the rest of the management team to determine the financial impact and trade-off of possible product, pricing and capacity building strategies.
- Financial statement analysis and preparation of profitability summary reports, re: products, distribution centers, factories and the business as a whole.
- Work with marketing, production and business development as needed to help them estimate the financial impacts of their decisions.

COO

- Design a supply chain that delivers a sustainable advantage.
- Manage inventory throughout the supply chain.
- Monitor inventory levels and make sure that inventory is in the right place at the right time.
- Manage shipping agreements and all of its attributes, such as batch sizes, shipping mechanisms, and reorder points.
- Anticipate inventory buildup and stock outs and advise management team accordingly.
- Optimize production capabilities for given production technologies and work in conjunction with the marketing and financial leaders of the organization to make choices about capacity building decisions.
- Optimize distribution.
- Calculate unit production costs for products using different technologies and levels of capital expenditures and recurring costs.
- Provide marketing team with cost forecasts aligned with their demand forecasts.
- Determine the financial and strategic trade-offs of building for efficiency, flexibility, fast response, etc.
- Manage short term, medium term and long capacity through order quantities, recurring expenditure levels, and capital expenditures.
- Work with the CFO to keep track of profitability at the factory and distribution center levels.
- Track and manage idle time in the factories

CMO

- Develop pricing strategies.
- Estimate demand and price elasticity by performing market research and analyzing sales data for your firm and competitor firms.
- Work with the CSO to research competitors' products and prices.
- Design new products and manage the new product pipeline.
- Set sales volume targets and prices for new and old products.
- Work closely with the production team to keep them informed of changes in demand and new product development and understand when inventory is building up and determine whether it needs to be marked down.
- Work with the CFO to evaluate the financial aspects of product development.
- Manage product portfolio.

CSO

- Business Development
- Work with the CEO and CMO to develop distribution strategies.
- Work with the CFO to determine contract provisions and terms and execute the contracts.
- Negotiate contracts with other teams to manage product distribution and establish distribution channels.
- Monitor what other teams are doing in terms of strategies, products, investments and report back to the rest of the management team.